

CSA1717-ARTIFICIAL INTELLIGENCE PRACTICAL LAB EXPERIMENTS

TOWERS OF HANOI

Aim:

To implement the Towers of Hanoi problem using recursion in Prolog.

Objective:

To demonstrate recursive problem solving by moving N disks from one peg to another while following the rules of Towers of Hanoi.

Algorithm: Start.

If there is one disk, move it directly. For N disks:

- **Move N-1 disks from source to auxiliary.**
- **Move the largest disk from source to destination.**
- **Move N-1 disks from auxiliary to destination. Display each movement.**

Stop.

Flowchart:

START

| v

Read N

| v

Is N = 1?

/ \

Yes No

||

Move Disk Move N-1

| Source->Aux

||

| Move Largest

||

| Move N-1

| Aux->Dest

\\

v v

Display Moves

| v

STOP

Prolog Program:

% Planets Database

planet(mercury, terrestrial, 1).

planet(venus, terrestrial, 2).

planet(earth, terrestrial, 3).

planet(mars, terrestrial, 4).

planet(jupiter, gas_giant, 5).

planet(saturn, gas_giant, 6).

planet(uranus, ice_giant, 7).

planet(neptune, ice_giant, 8).

terrestrial(P) :-

planet(P, terrestrial, _).

gas_giant(P) :-

planet(P, gas_giant, _).

ice_giant(P) :-

planet(P, ice_giant, _).

Output:

Result:

Thus, the Towers of Hanoi problem is successfully implemented using recursion in Prolog.